



ABES Engineering College, Ghaziabad

Department of Applied Sciences & Humanities

Session: 2025-26

Year/ Semester: I/2nd

Branch/Section: CSE21-30 / AIML / ELCE/ME

Course Code: 25AS201

Course Name: Applied Physics

Assignment: 01

Date of Assignment:

Date of Submission:

MM:10 marks

Q. No.	Question	CO	KL	Marks
1	State reasons for need of quantum mechanics?	CO1	K1	1
2	Compare the wavelength of a photon and an electron, if they have same momentum.	CO1	K1	1
3	What is the difference between matter waves and electromagnetic waves?	CO1	K1	1
4	Define wave packet.	CO1	K1	1
5	(i) What is the physical significance of wave function? (ii) Give the essential conditions or requirements that a wave function must fulfil.	CO1	K1	1
6	Define the eigen value and eigen function.	CO1	K1	1
7	Why is tunneling not possible in classical mechanics?	CO1	K1	1
8	An electron and a proton of same energy approaches a barrier potential whose height $V > E$, which have larger probability to be at the other side of the barrier.	CO1	K1	1
9	(i) What are the fundamental principles of quantum computing? (ii) Define quantum bit in quantum computing.	CO1	K1	1
10	Mention advantages of quantum cryptography over ordinary cryptography.	CO1	K1	1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



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Tutorial /Practice sheet : 01

Q. No.	Question	CO	KL	
1	(i) A proton is moving with a speed of 2×10^8 m/sec. Find the wavelength of wave associated with it. (1.47×10^{-5} Å) (ii) Calculate the velocity and kinetic energy of a neutron having de-Broglie wavelength 1Å. (K.E = 0.082ev)	CO1 CO1	K3 K3	
2	(i) Calculate the de-Broglie wavelength of an α particle accelerated through a potential difference of 200 volts. (0.00716 Å) (ii) Calculate the de- Broglie wavelength associated with a proton moving with a velocity equal to $(1/20)^{\text{th}}$ velocity of light. (2.643×10^{-14} m.)	CO1 CO1	K3 K3	
3	(i) Determine the probability of finding a particle trapped in a box of length L in the region from 0.45L to 0.55L for the ground state. (19.8%) (ii) A particle confined to move along x-axis has the wave function $\psi = ax$ between $x=0$ and $x=1$ and $\psi = 0$ elsewhere. Find the probability that the particle can be found between $x=0.35$ and $x=0.45$. P = $0.0161a^2$	CO1 CO1	K3 K3	
4	(i) An electron is bound in one dimensional box which has width 2.5Å. Assuming the height of the box to be infinite, calculate two lowest permitted energy values of the electron. (6.04ev, 24.16ev) (ii) An electron is trapped in one-dimension region of length 1Å. Find the amount of energy that must be supplied to excite the electron from ground state to first excited state. (113.07ev)	CO1 CO1	K3 K3	
5	(i) A 12.0-eV electron encounters a barrier of height 15.0 eV. If the probability of the electron tunneling through the barrier is 2.5 %, find its width. (0.21nm) (ii) Electrons with energies of 1.0 ev are incident on a barrier 3 ev high and 0.2 nm wide. find the approximate probability for the electrons to penetrate the barrier. (0.084)	CO1 CO1	K3 K3	

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